Mean Reversion in OIS-Treasury Spreads: Evidence, Regime Dependence, and a Hidden Markov Model Trading Filter

Chase Leibowitz (crl65), Dhruv Trivedi (dpt10)
Jose Maldonado (jam348), Felipe Sanchez-Noguera (fs172)

2026-04-19

1 Introduction

When interest-rate swaps and Treasury bonds both price the same thing — where short-term rates are headed — their yields should stay close together. The gap between them (the "spread") tends to hover around a long-run average, compressing in calm markets and widening under stress. That mean-reverting behavior underpins a classic fixed-income strategy: when the spread drifts too far, bet on it snapping back. These spreads affect pension funding ratios, corporate hedging costs, and Federal Reserve market-function assessments, making them a subject of broad practical and policy relevance.

This report tests that idea systematically using OIS swap rates and Treasury yields at 2Y, 5Y, and 10Y maturities, covering 1,890 daily observations from 2019 to 2026. We address three research questions and one applied extension:

- **RQ1** — Do OIS-Treasury spreads mean-revert after extreme $\pm 2\sigma$ signals?
- **RQ2** — Is this result robust to the choice of rolling window, threshold, and forward horizon?
- **RQ3** — Does mean reversion differ across rate-direction and volatility regimes?
- **Applied extension** — Can an unsupervised Hidden Markov Model detect those regimes in a way that improves out-of-sample trading performance more than a supervised classifier?

2 Key Terms

Basis points (bps) 1/100 of a percentage point; 100 bps = 1 pp.

Treasury yield / CMT Daily yield on U.S. government bonds at standardized maturities (2Y, 5Y, 10Y), published by the Federal Reserve.

SOFR The rate banks pay to borrow overnight against Treasury collateral; the benchmark floating rate for USD OIS contracts.

OIS (Overnight Index Swap) A swap where one party pays a fixed rate and receives compounded SOFR; no principal changes hands, so credit risk is minimal.

EFFR (Effective Fed Funds Rate) The overnight interbank lending rate set by the Fed; used here as a rate-regime classifier, not the OIS reference rate.

OIS-Treasury Spread $s_{t,\tau} = \mathrm{OIS}_{t,\tau} - \mathrm{DGS}_{t,\tau}$; negative when OIS yields less than Treasuries.

Liquidity premium Extra yield investors demand for less-liquid assets; Treasuries' superior liquidity bids their yields below OIS.

Relative value (RV) A trade that bets on a mispricing between related instruments correcting, not on market direction.

Mean reversion The tendency of a series to return to its average after a large deviation.

MOVE Index Market-implied volatility of Treasury yields — the rates equivalent of the VIX.

HMM (Hidden Markov Model) A model that infers unobserved market "states" from data without hand-labeling; parameters estimated via Baum-Welch (EM), states decoded via Viterbi.

Sharpe ratio Annualized return ÷ annualized return volatility; higher is better.

Maximum drawdown Largest peak-to-trough loss in cumulative P&L.

Calmar ratio Annualized return ÷ maximum drawdown; penalizes deep losses more than Sharpe.

Cohen's d Standardized effect size (mean difference ÷ pooled std); shows practical, not just statistical, significance.

FDR (False Discovery Rate) Fraction of rejected hypotheses that are false positives; controlled via Benjamini-Hochberg when many tests are run simultaneously.

3 Data

3.1 Sources

Source	Ticker / Series	Description	Role
FRED	DGS2, DGS5, DGS10	Treasury CMT yields, 2/5/10Y	Benchmark yields
FRED	EFFR	Effective Fed Funds Rate	Rate regime classifier
LSEG	USDOIS2/5/10Y=PYNY	SOFR OIS par rates, 2/5/10Y	Swap rates
LSEG	.MOVE	ICE BofA MOVE Index	Implied Treasury vol

3.2 Cleaning and Construction

FRED holiday sentinels (`"."`) were converted to `NaN`; LSEG OIS CSVs were parsed by locating the `Date` header dynamically, with mid-rates computed as (Bid + Ask)/2. Both panels were merged by date with an inner join and a 5-business-day forward-fill for federal holidays. Spreads are $s_{t,\tau} = \mathrm{OIS}_{t,\tau} - \mathrm{DGS}_{t,\tau}$ and the trading signal is the 60-day rolling z-score $z_{t,\tau} = (s_{t,\tau} - \bar{s}^{(60)}_{t,\tau})/\sigma^{(60)}_{t,\tau}$. The final panel has 1,890 daily observations (2019-01-01 to 2026-04-14). Full pipeline details are in Appendix A. **2Y outliers:** 23 dates show `spread_2y > 1.0 pp` from an LSEG quoting artifact (OIS jumps ~200 bp while DGS2 stays flat then reverts). These are kept in classical analysis but excluded from all ML features.

3.3 Summary Statistics

Column	Mean (pp)	Std (pp)	Min	Max	% Negative
spread_2y	−0.093	0.258	−0.352	2.149	82.3%
spread_5y	−0.247	0.100	−0.470	0.463	99.9%
spread_10y	−0.336	0.133	−0.745	0.131	99.9%

OIS trades below Treasuries essentially 100% of the time at 5Y and 10Y, consistent with a positive Treasury liquidity premium.

4 Modules

Module 3 — Visualization *(data investigation, analysis, final report).* Time-series plots, histograms, event-study curves, heatmaps, regression scatters, and cumulative P&L curves (all course-covered) anchor every result. We also produced SHAP summary plots — not covered in class — which attribute individual model predictions to specific features, adding interpretability beyond standard feature-importance charts.

Module 4 — Data Wrangling *(data gathering, cleaning).* LSEG CSV parsing with dynamic header detection, multi-source inner join with holiday forward-fill, 60-day rolling z-score and MOVE z-score construction, EFFR-change regime labeling, non-overlapping signal event extraction, and lagged-feature engineering. The OIS data required custom parsing logic before it could be merged with FRED.

Module 6 — Combining Data *(data gathering, analysis).* Merged FRED Treasury yields, LSEG OIS par rates, and the MOVE index into a single daily panel; attached regime labels and HMM state assignments; built the event-level feature matrix for the classifier.

Module 7 — Statistical Inference *(data analysis).* One-sample t-tests and percentile bootstrap CIs (course-covered) anchor every RQ. We supplemented through independent research: the Wilcoxon signed-rank test (non-parametric, no normality assumption), Cohen's d (effect size), Welch two-sample t-tests (unequal variances), Benjamini-Hochberg FDR correction (controls false positives across 162 robustness cells), and two-way ANOVA with partial η^2 (decomposes variance across two regime factors and their interaction). Each was chosen as the standard tool for its inferential task.

Module 8 — Prediction and ML *(data analysis).* Supervised classification (course-covered) was applied via Random Forest with a Logistic Regression baseline. The HMM, Baum-Welch/Viterbi algorithms, feature ablation, and walk-forward backtest came from prior work and independent study — the most appropriate tools for testing whether an unsupervised regime model improves trading performance.

5 Results and Methods

Code and data: `github.com/crl65/COMPSCI216-Project`.

5.1 RQ1: Mean Reversion After Extreme Z-Scores

For each of 3 maturities $\times$ 2 signal directions $\times$ 3 forward horizons, we extracted non-overlapping $|z| > 2$ events (minimum 5-day gap) and tested whether the subsequent spread change is systematically in the direction predicted by mean reversion. We used t-tests and bootstrap CIs (course-covered), supplemented with the Wilcoxon signed-rank test, Cohen's d, and BH-FDR correction. Table 1 shows 10-day results; the full 18-cell table is in Appendix B.

Maturity	Signal	n	Mean (bps)	Cohen's d	t-stat	p_{BH}	Sig
2Y	z< −2	33	+2.69	+0.48	+2.76	0.019	∗
2Y	z> +2	40	−21.17	−0.37	−2.35	0.039	∗
5Y	z< −2	48	+2.68	+0.87	+6.02	< 0.001	∗ ∗ ∗
5Y	z> +2	37	−3.85	−0.33	−2.00	0.055	.
10Y	z< −2	44	+2.94	+0.53	+3.53	0.003	∗∗
10Y	z> +2	37	−4.10	−0.63	−3.85	0.002	∗∗

: **Table 1.** RQ1 results, 10-day horizon. All 18 cells show the predicted sign; 12 of 18 survive BH correction. Effect sizes at 5Y ($d \approx 0.87$) are large by conventional standards — the signal is practically meaningful, not just statistically detectable. Codes: $* * *$ $p < 0.001$, $**$ $p < 0.01$, $*$ $p < 0.05$.

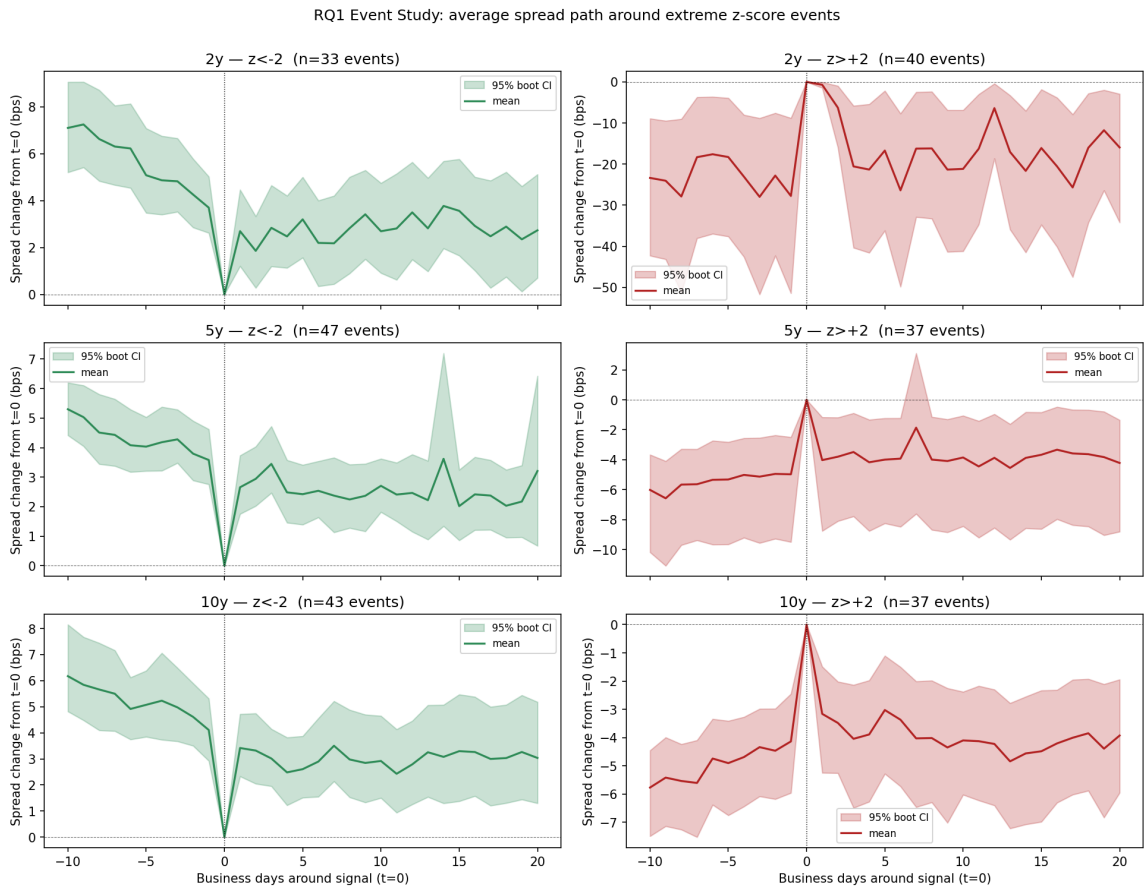

Figure 1: Each panel shows the average spread path in a 30-day window centered on the signal date, normalized to zero at signal time. Rows = maturities; columns = signal direction (spread too low vs. too high). In every panel the spread drifts back toward zero — the expected mean-reversion shape, roughly half-complete by day 10.

At 5Y $z < -2$ (10 days), mean reversion is +2.68 bps (Cohen's $d = 0.87$, CI $[+1.81, +3.61]$) — a large, consistent effect that exceeds the 1 bp round-trip cost used in the backtest and is therefore tradable in practice.

5.2 RQ2: Robustness Across Parameter Grids

Across 162 combinations of windows (30–120 days), thresholds (1.5–2.5σ), and horizons (5–20 days) — with BH-FDR applied throughout — the direction never flips once and the magnitude barely moves (Table 2). This rules out parameter selection as an explanation: mean reversion in OIS-Treasury spreads is a structural feature, not a tuning artifact. The p-value heatmap is in Appendix B.

Quantity	Value
Total combinations	162
Combinations with sign predicted by reversion	162 (100%)
Combinations surviving BH FDR at $q < 0.05$	118 (73%)
Range across *window* levels	2.83 bps
Range across *threshold* levels	2.72 bps
Range across *horizon* levels	0.64 bps

: **Table 2.** RQ2 stability across 162 parameter combinations.

5.3 RQ3: Regime Dependence

To test whether the magnitude of mean reversion depends on the macro environment, we split events by rate direction (rising vs. falling EFFR over 60 days) and volatility level (MOVE above/below its 60-day median), then applied two-way ANOVA (see Modules). The answer is unambiguous: rate regime drives reversion size; volatility does not. The continuous MOVE relationship is in Appendix B; Table 3 gives the ANOVA decomposition.

Maturity	Signal	Term	F	p	partial η^2
5Y	z< -2	rate_regime	17.26	< 0.001	0.282
5Y	z< -2	vol_regime	0.01	0.930	0.000
5Y	z< -2	rate $\times$ vol	0.78	0.383	0.017
5Y	z$> +2$	rate_regime	6.59	0.015	0.167
5Y	z$> +2$	vol_regime	0.42	0.521	0.013
5Y	z$> +2$	rate $\times$ vol	2.16	0.151	0.062

: **Table 3.** Two-way ANOVA for 5Y signals. Rate direction alone explains 28% of variance in reversion size ($\eta^2 = 0.28$, $p < 0.001$) — a large effect implying spreads are under fundamentally different tension when rates are climbing. Volatility contributes nothing ($\eta^2 \approx 0$, $p = 0.93$). Rising-rate periods produce 3$\times$ larger reversion (+5.40 vs +1.67 bps; Appendix B), which is the actionable signal for a practitioner.

5.4 Applied Extension: HMM, Random Forest, and Backtest

We fit a 2-state Gaussian HMM on (`spread_5y`, 60-day EFFR change). The two states separate cleanly into recognizable macro episodes: a "stress" state (higher MOVE, 48% rising-rate days) and a "stable" state (lower MOVE, 18% rising-rate), each persisting ~6 months — slow structural shifts that a fixed-income desk could actually condition on. Look-ahead bias was ruled out via pre-2024 refit: 100% of OOS state assignments matched (Appendix C).

A Random Forest (500 trees, depth 4, 10 features) was trained to predict 5Y signal reversion ≥ 3 bps within 10 days. It achieves AUC 0.544 on 29 OOS events — barely above chance — and feature ablation (Appendix C) shows adding `hmm_state` *lowers* AUC by 0.12. The failure is structural: 56 training events and a 69% majority class leave the RF no choice but to memorize the base rate. It cannot recover latent regime structure from sparse outcome labels the way the HMM does by fitting directly to market dynamics.

The backtest ran from 2024-01-02 to 2026-03-24 (29 events, 1 bp round-trip cost, exit at 10 days or z-crossing), comparing three strategies:

Strategy	Trades	Win %	Mean (bps)	Total (bps)	Sharpe	Max DD (bps)	Calmar
1 — Rules-based	20	55.0	+0.82	+16.4	0.39	−9.6	0.73
2 — RF-filtered	20	55.0	+0.82	+16.4	0.39	−9.6	0.73
3 — HMM-filtered (S=1)	11	72.7	+1.55	+17.1	0.51	−9.4	0.78

: **Table 4.** Out-of-sample strategy comparison, 2024-01-02 through 2026-03-24.

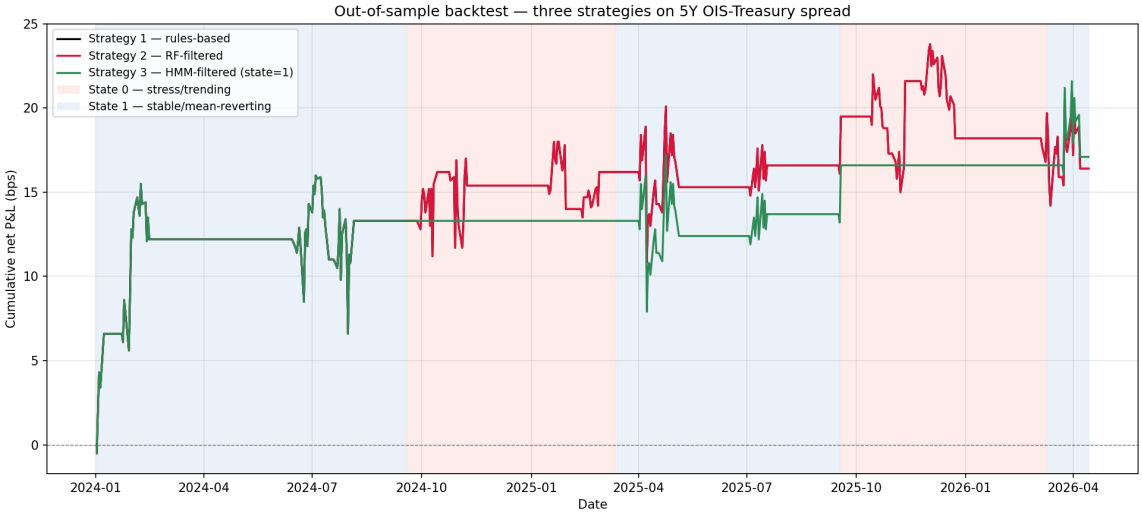

Figure 2: Cumulative P&L for all three strategies. Background shading shows the HMM regime: red = stress (State 0), blue = stable (State 1). The HMM-filtered strategy (green) skips trades during stress periods, avoiding most losses, and ends with a higher Sharpe ratio while taking only 55% as many trades. Full trade log in Appendix D.

The RF filter is inoperative: 28 of 29 events clear its threshold, making Strategy 2 identical to Strategy 1 in practice. The HMM filter is the opposite — it identifies that all 9 stress-state trades produced every dollar of drawdown and zero net P&L (Appendix D), and simply skips them. The result: Sharpe rises from 0.39 to 0.51, win rate from 55% to 73%, without changing the underlying entry signal at all.

Central finding. The same `hmm_state` that *harmed* the RF as a continuous feature *helped* the backtest as a binary gate — how you represent a regime signal determines whether you capture it.

6 Limitations and Future Work

The OOS window (27 months, 29 events) is too short for statistically conclusive strategy comparisons; the 85-event ML training set is tight for a 10-feature classifier. The 2Y spread retains 23 LSEG quoting artifacts; transaction costs are stylized at 1 bp flat. **Future work** could extend history to 2010 via SOFR/LIBOR-OIS splicing, add more maturities, replace the batch HMM with a real-time filter, and replicate in EUR and GBP. At scale, systematic RV strategies risk amplifying dislocations during the stress windows when they trade most — worth monitoring in any live deployment.

7 Conclusion

We find robust mean reversion in OIS-Treasury spreads across three maturities, 1,890 daily observations, and 162 parameter combinations — 100% of cells carry the predicted sign and 73% survive FDR correction (RQ1, RQ2). Rate regime is the dominant moderator: rising-rate environments roughly triple the 5Y reversion magnitude, with partial $\eta^2 = 0.28$; volatility regime matters continuously in the MOVE scatter but not as a binary split (RQ3). In the applied extension, a 2-state HMM used as a binary trade gate raises out-of-sample Sharpe from 0.39 to 0.51 and win rate from 55% to 73%, while the identical regime signal fed into a Random Forest actively hurts classifier AUC. The lesson is representation: given scarce labels, imposing known regime structure beats trying to learn it.

8 Collaboration Reflection

Alignment with plan. Our collaboration plan called for divided ownership across data, inference, and ML — that structure held throughout. The biggest deviation was a time-consuming mid-project data pivot to real LSEG OIS par rates.

What went well / not well. Versioned outputs in `outputs/` made handoffs frictionless. What did not go well: we underestimated the data-pivot cost and initially skipped BH-FDR correction, requiring a mid-project rerun of all inference.

What we'd do differently. Start with real swap data, investigate more advanced statistical inference methods earlier to save time, further split responsibilities between people so the same people aren't working on the same task.

Sharing plan. All code, data, and figures are in the shared GitHub repository (`github.com/crl65/COMPSCI216-Project`).. We plan to keep the repository public.

9 AI Disclosure

Artificial Intelligence Tools: ChatGPT 5.2 via ChatGPT.com. Artificial Intelligence Tools: Claude Opus 4.6 via Claude Code. Artificial Intelligence Tools: Claude Sonnet 4.5 5.2 Claude Code. Methodology: ChatGPT was used for determining EDA methodology given the dataset. Methodology: ChatGPT and Claude were used for designing the research framework and for analyzing RQs. Methodology: Claude was used to aid in researching and understanding more advanced statistical inference concepts than those covered in class. Methodology: Claude Code was used for code syntax execution. Writing – Review and Editing: Claude Code (within Quarto) was used for assistance in writing and editing.

10 Appendix A — Full Data Pipeline

10.1 Full cleaning methodology

The FRED Treasury series returns the sentinel `"."` on U.S. federal holidays and other closed dates; these were coerced to `NaN` before any numerical operation. The LSEG OIS CSVs have a non-standard multi-row header that varies slightly across the 2Y, 5Y, and 10Y files; our loader locates the `Date` header row dynamically, drops preceding metadata, parses the `Bid` and `Ask` columns as floats, and computes the mid-rate as $(\text{Bid} + \text{Ask})/2$. Scale was verified manually against Bloomberg snapshots on four random dates to confirm percentage-point units (sample max = 6.31 pp at 2Y, October 2023).

The Treasury and OIS panels were merged by date with an **inner join** so every retained row has both a DGSx and an OIS_Xy observation. Forward-fills with a five-business-day cap were applied to cover U.S. federal holidays where the Treasury market is closed but the OIS market trades (Columbus Day, Veterans Day, etc.). The MOVE index was loaded from its own LSEG CSV and left-joined onto the spread panel so that occasional missing MOVE dates do not drop spread observations.

10.2 2Y outlier investigation

A mechanical scan for `spread_2y > 1.0 pp` returned 23 observations, clustered in April-October 2022. On each such date the LSEG OIS 2Y rate jumps roughly 200 basis points relative to its prior value while the matched DGS2 stays smooth, and the OIS quote reverts on the following business day. We interpret these as LSEG quoting artifacts rather than real market events. They are retained in classical analysis (their inclusion is conservative — they dilute the 2Y reversion signal) and excluded from the ML feature set.

10.3 Supplementary EDA figures

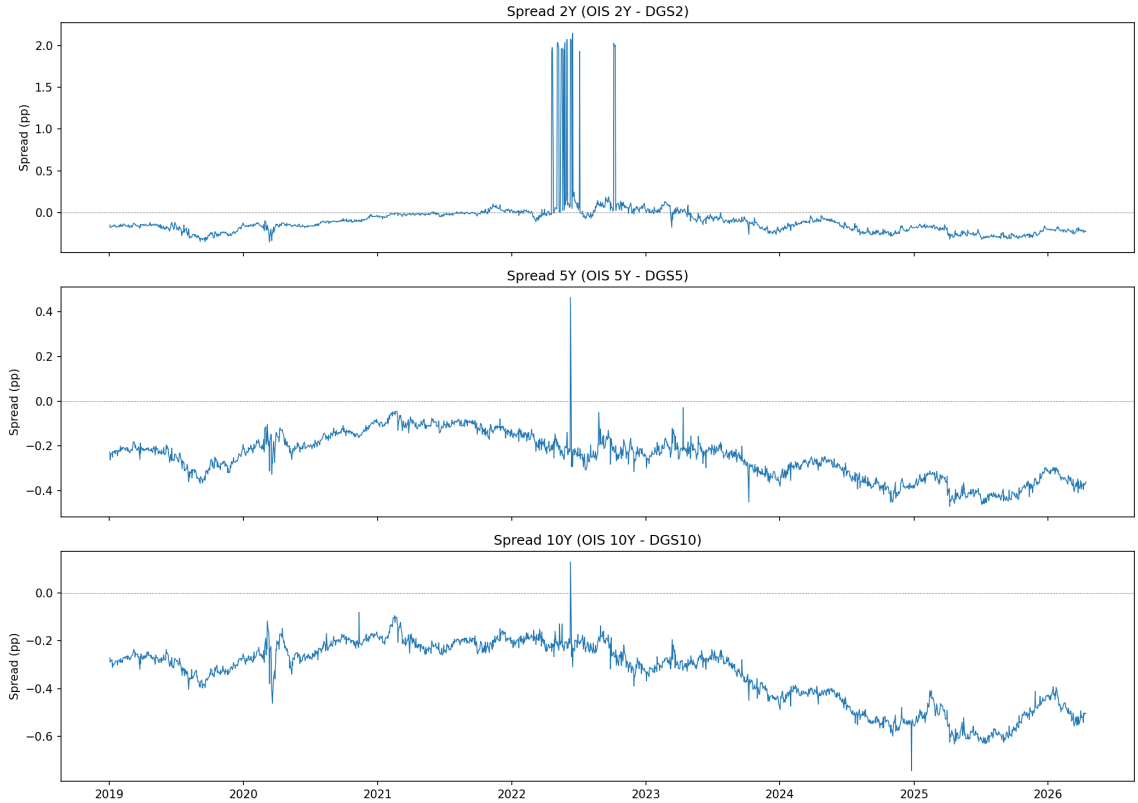

Figure 3: Spread time series (OIS minus matched-maturity Treasury). The 2Y artifacts of 2022 are visible as isolated upward spikes.

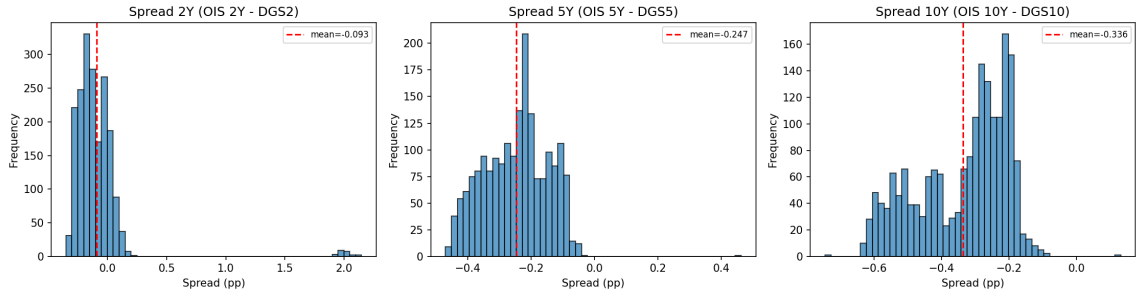

Figure 4: Spread distributions by maturity. Red dashed lines mark sample means. The 2Y right tail is dominated by LSEG quoting artifacts.

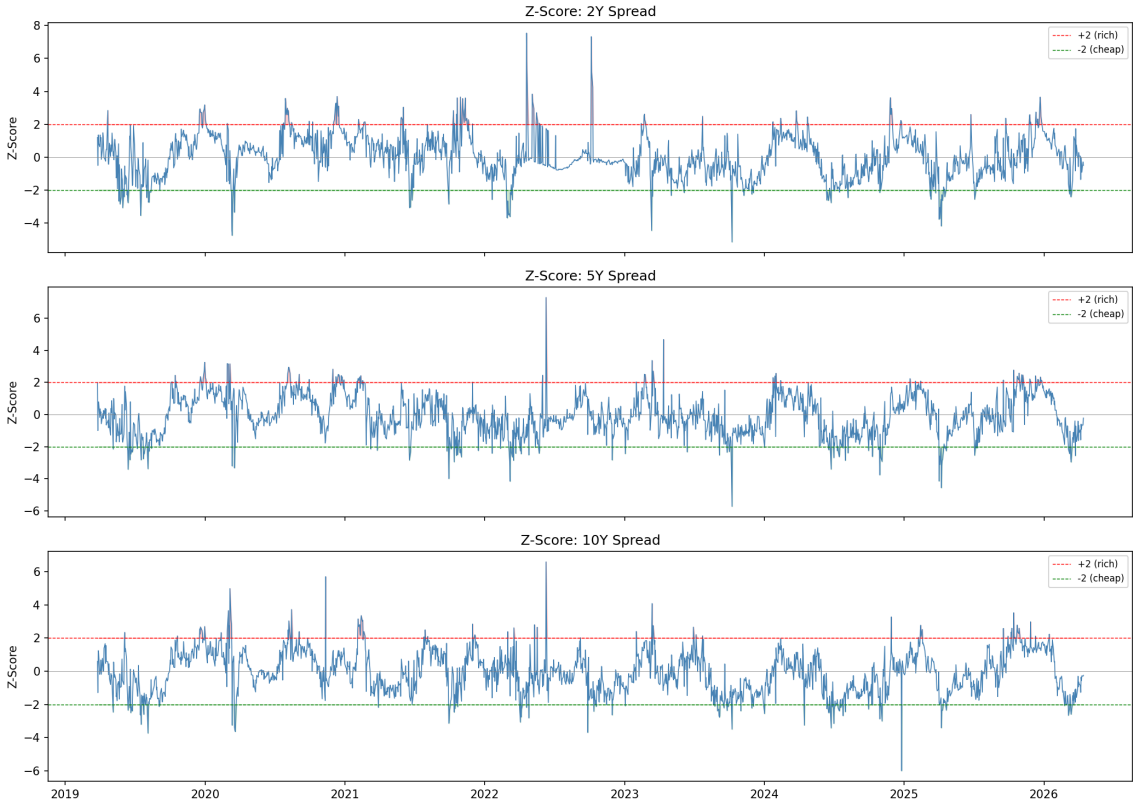

Figure 5: 60-day rolling z-scores with ± 2 thresholds marked. Breach rates are consistent with a mean-reverting series with moderately fat tails.

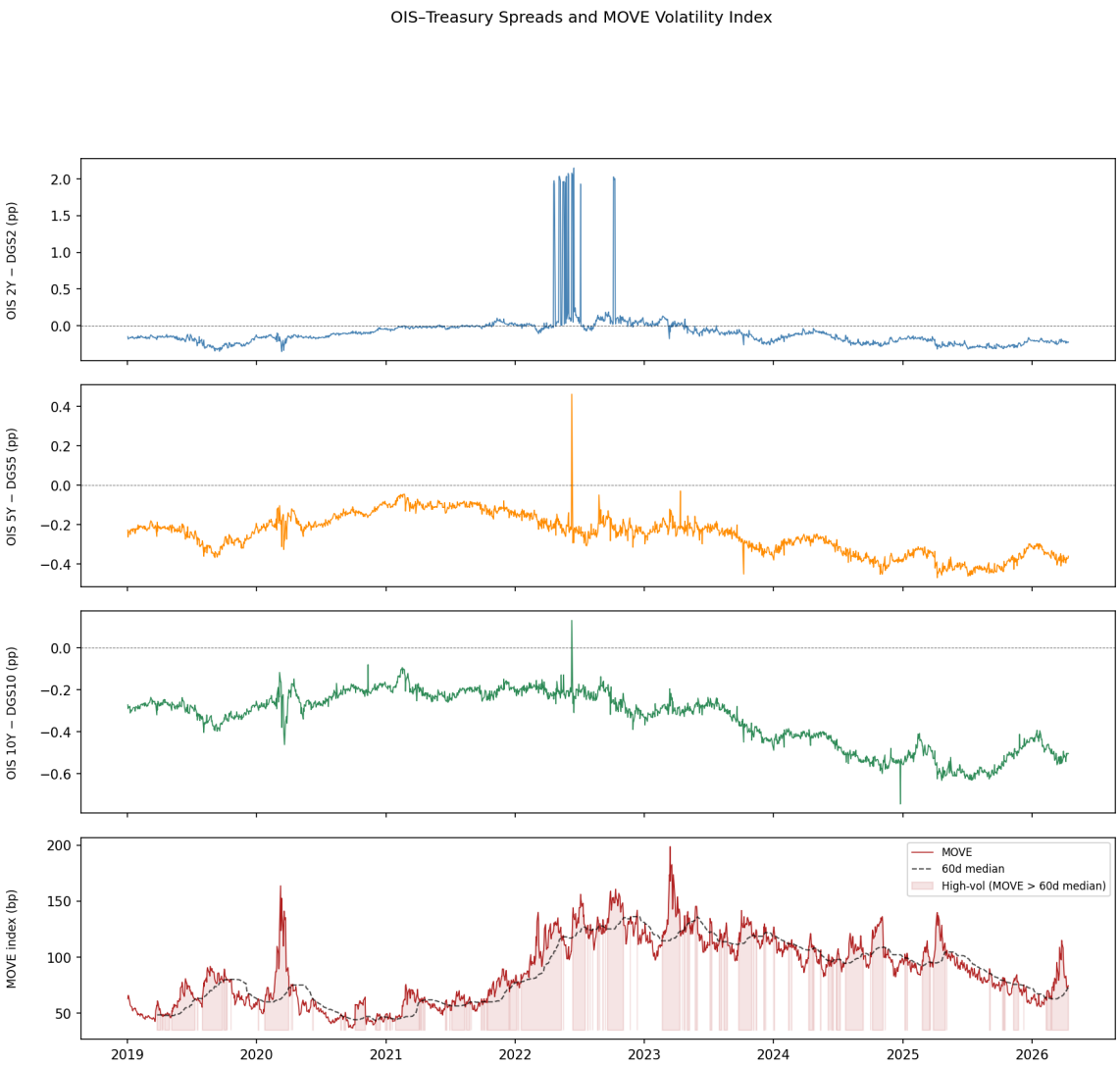

Figure 6: Overview panel: three spreads stacked with the MOVE index. Shaded regions mark high-volatility episodes (MOVE above rolling 60-day median).

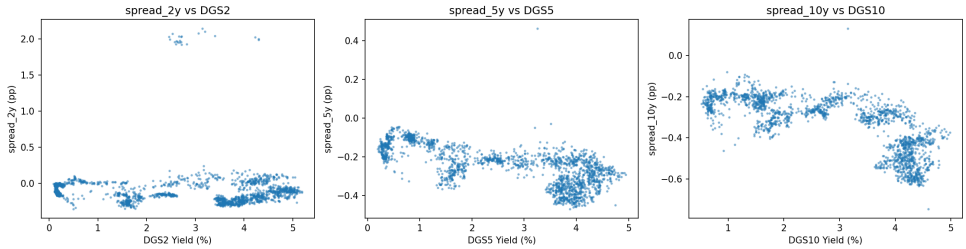

Figure 7: Spread-vs-yield scatter: OIS rates vs. matched-maturity Treasuries. The line-of-best-fit confirms the systematic negative offset.

Figure 8: Pairwise correlation matrix across spreads, yields, MOVE, and rolling z-scores.

11 Appendix B — Extended Statistical Results

11.1 Full RQ1 master table (all horizons)

Maturity	Signal	Horizon	n	Mean (bps)	Cohen's d	t-stat	p_{BH}	Sig
2Y	z< −2	5	33	+1.41	+0.28	+1.60	0.138	.
2Y	z< −2	10	33	+2.69	+0.48	+2.76	0.019	*
2Y	z< −2	20	33	+4.15	+0.61	+3.51	0.005	**
2Y	z> +2	5	40	−15.83	−0.28	−1.78	0.112	.
2Y	z> +2	10	40	−21.17	−0.37	−2.35	0.039	*
2Y	z> +2	20	40	−24.86	−0.43	−2.73	0.020	*
5Y	z< −2	5	48	+1.64	+0.64	+4.42	< 0.001	***
5Y	z< −2	10	48	+2.68	+0.87	+6.02	< 0.001	***
5Y	z< −2	20	48	+3.42	+0.83	+5.75	< 0.001	***
5Y	z> +2	5	37	−2.17	−0.22	−1.34	0.211	.
5Y	z> +2	10	37	−3.85	−0.33	−2.00	0.055	.
5Y	z> +2	20	37	−5.14	−0.36	−2.19	0.042	*
10Y	z< −2	5	44	+1.49	+0.36	+2.37	0.032	*
10Y	z< −2	10	44	+2.94	+0.53	+3.53	0.003	**
10Y	z< −2	20	44	+4.11	+0.58	+3.83	0.002	**
10Y	z> +2	5	37	−2.52	−0.50	−3.02	0.008	**
10Y	z> +2	10	37	−4.10	−0.63	−3.85	0.002	**
10Y	z> +2	20	37	−5.47	−0.70	−4.24	< 0.001	***

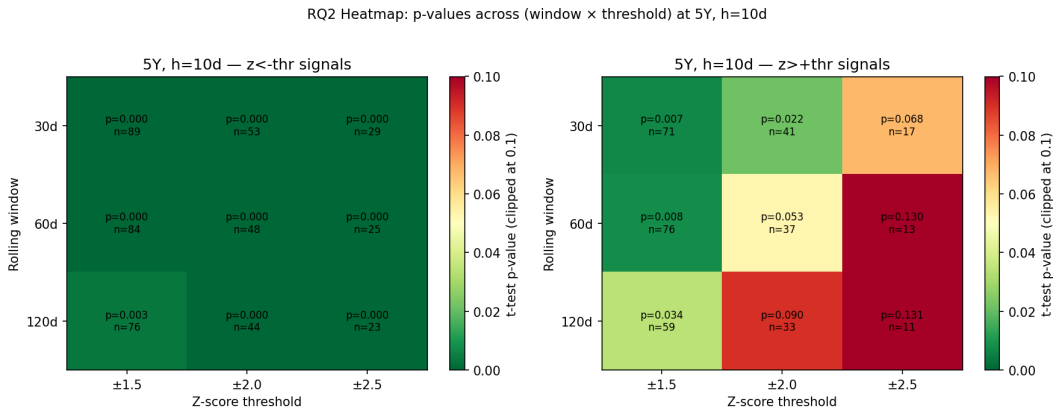

Figure 9: P-values for the 5Y spread at the 10-day horizon across all window × threshold combinations. Green = significant. All nine cells point in the same direction — the result does not depend on parameter choice.

11.2 RQ2 p-value heatmap

11.3 RQ2 coefficient plot

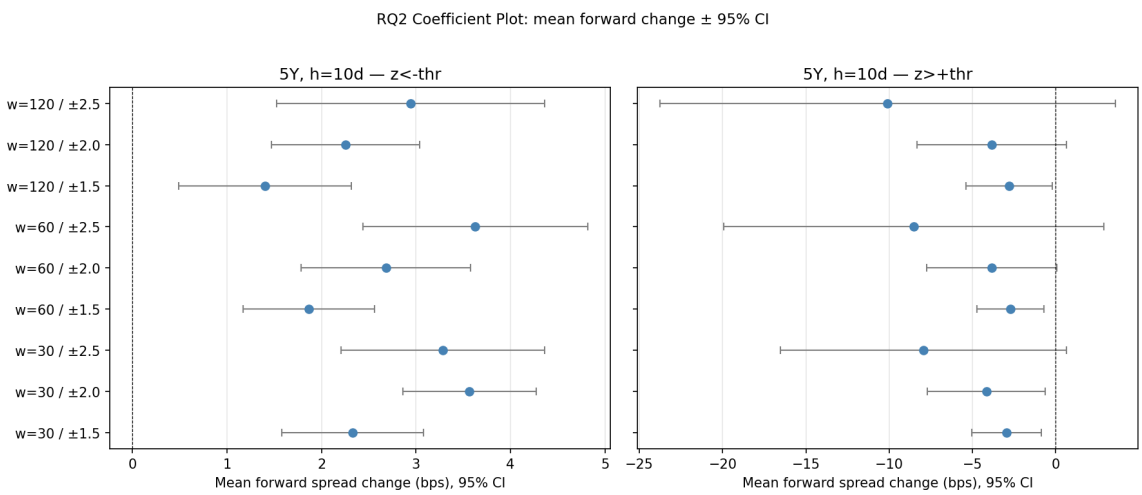

Figure 10: RQ2 coefficient plot for the 5Y spread at the 10-day horizon. Points are mean forward changes in bps; bars are 95% CIs. Sign and magnitude are stable across all nine (window, threshold) combinations.

11.4 RQ3 Part A — Rate-regime split

Maturity	Signal	Rising (bps, n)	Falling (bps, n)	p-diff	Cohen's d
5Y	z< −2	+5.40 (13)	+1.67 (35)	0.001	1.21
5Y	z> +2	−10.84 (11)	−0.89 (26)	0.131	−0.85
10Y	z< −2	+3.11 (13)	+2.86 (31)	0.869	0.04
10Y	z> +2	−5.37 (15)	−3.23 (22)	0.371	−0.33

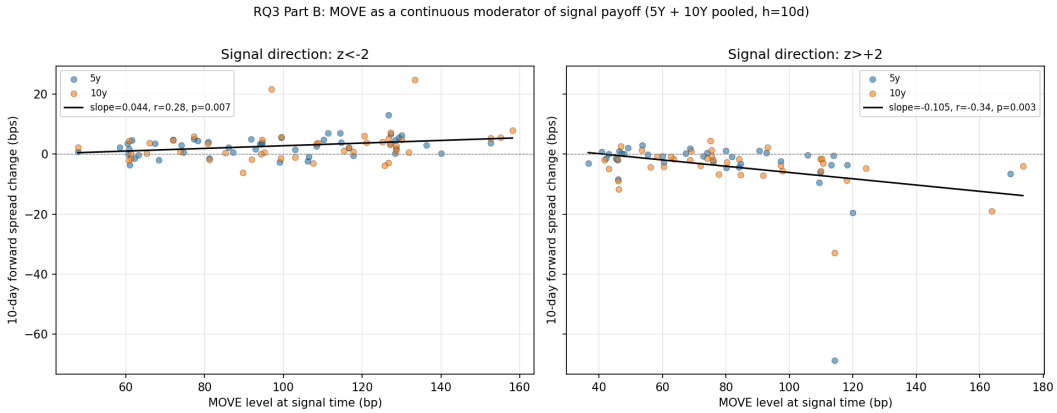

Figure 11: MOVE level at signal date vs. 10-day forward spread change, pooling 5Y and 10Y events. For positive-z signals (spread too wide), higher MOVE predicts a larger snap-back — OLS slope < 0, $r \approx -0.34$. A simple high/low split misses this; the continuous MOVE level is needed.

11.5 RQ3 Part B — MOVE scatter

11.6 RQ3 Part C — Joint heatmap

11.7 RQ3 Part D — Regime transition analysis (null finding)

A post-transition indicator (1 if regime changed within trailing 60 business days) was applied to both regimes. Vol-regime transitions produced degenerate splits (almost every date is within 60 days of a vol change). Rate-regime transitions produced clean splits but no significant post-transition effect ($p = 0.37$ at 5Y, $p = 0.52$ at 10Y). We report the null finding and note the indicator construction as the binding limitation.

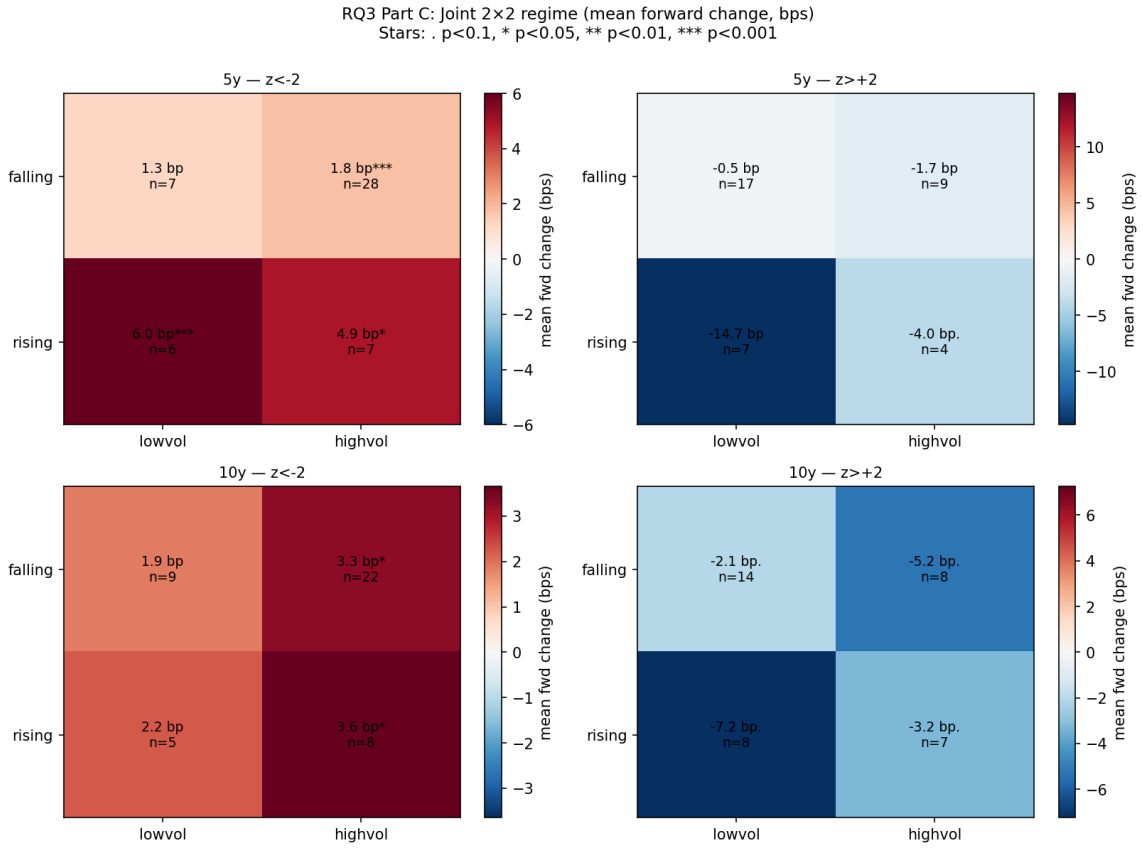

Figure 12: Joint 2$×2heatmap : mean10 − dayforwardchange(bps)ineachrate×$vol regime cell, by maturity and signal direction. Star annotations are raw t-test p-values.

12 Appendix C — ML Technical Details

12.1 HMM — state diagnostics

State	n	% Rising (EFFR)	Mean MOVE	Mean spread_5y (pp)
State 0 — *stress / trending*	856	48.1%	101.3	−0.269
State 1 — *stable / mean-reverting*	973	18.3%	77.6	−0.229

12.2 HMM — transition matrix

	To S0	To S1
From S0	0.994	0.006
From S1	0.005	0.995

Expected dwell times: ~171 days in S0, ~194 days in S1. Manual rate-regime agreement: 66%. Pre-2024-only refit produces 100% identical OOS state assignments (n = 591 days).

Figure 13: HMM regime decoding. Top: 5Y spread colored by Viterbi state. Bottom: MOVE colored by the same assignments. State 0 (red) concentrates in high-MOVE stress episodes.

12.3 Random Forest — metrics and baselines

Model	Accuracy	Precision	Recall	F1	ROC AUC
Random Forest (all features)	0.724	0.714	1.000	0.833	0.544
Logistic Regression	0.586	0.682	0.750	0.714	0.400
Naive (predict majority class)	0.690	0.690	1.000	0.816	0.500

12.4 HMM ablation

Feature set	Test AUC
Full feature set (10 features)	0.544
Full minus hmm_state (9 features)	0.667
ΔAUC (adding hmm_state)	−0.122

Including `hmm_state` as a continuous feature reduces AUC by 0.122. The regime signal is real but the RF cannot exploit it productively in a tree-split representation with only 56 training events.

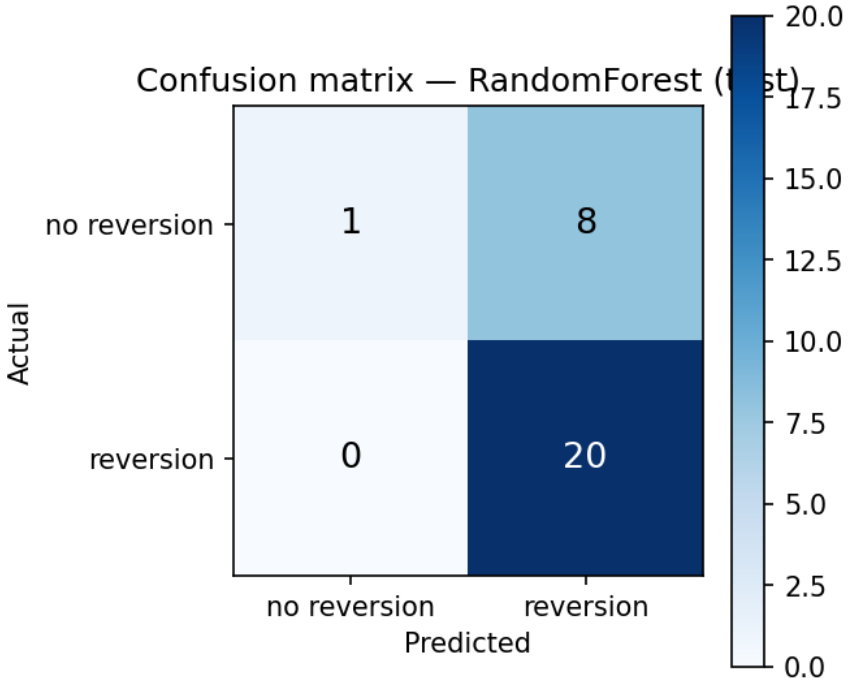

Figure 14: RF confusion matrix on 29 OOS events.

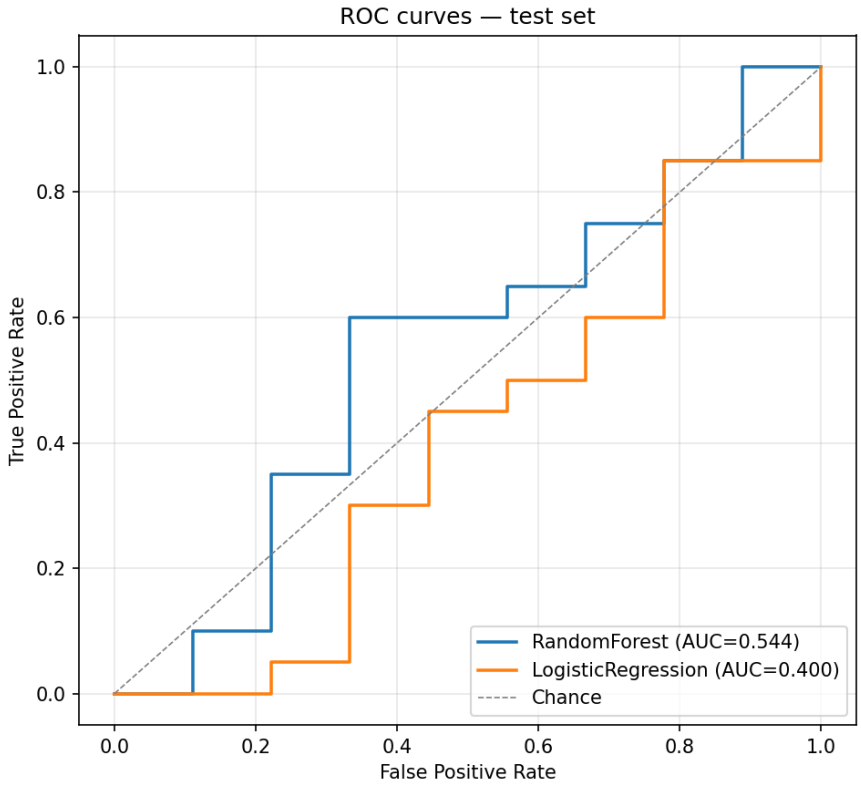

Figure 15: RF ROC curve on OOS events.

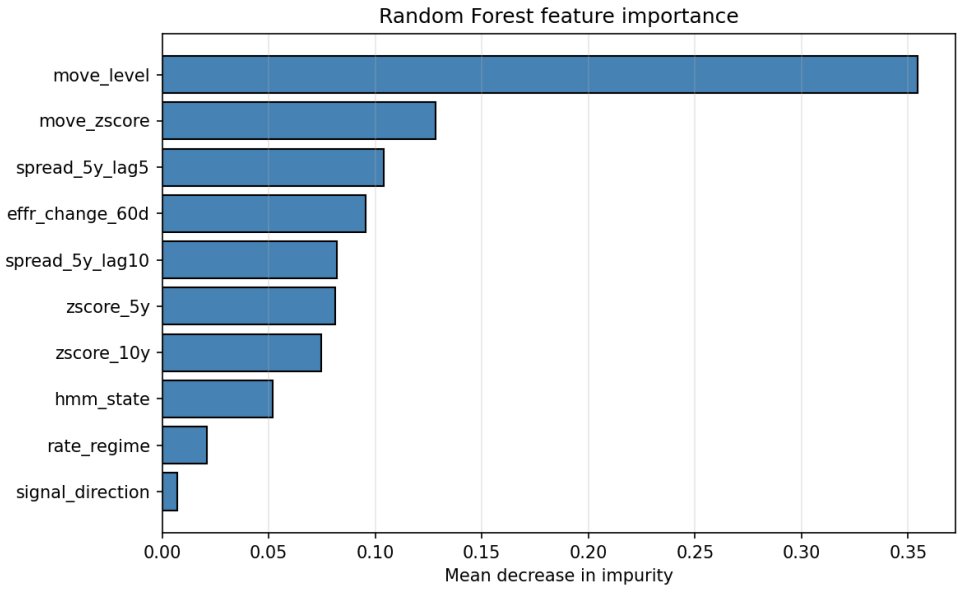

Figure 16: RF feature importance (mean decrease in impurity). MOVE level and its 60-day z-score dominate.

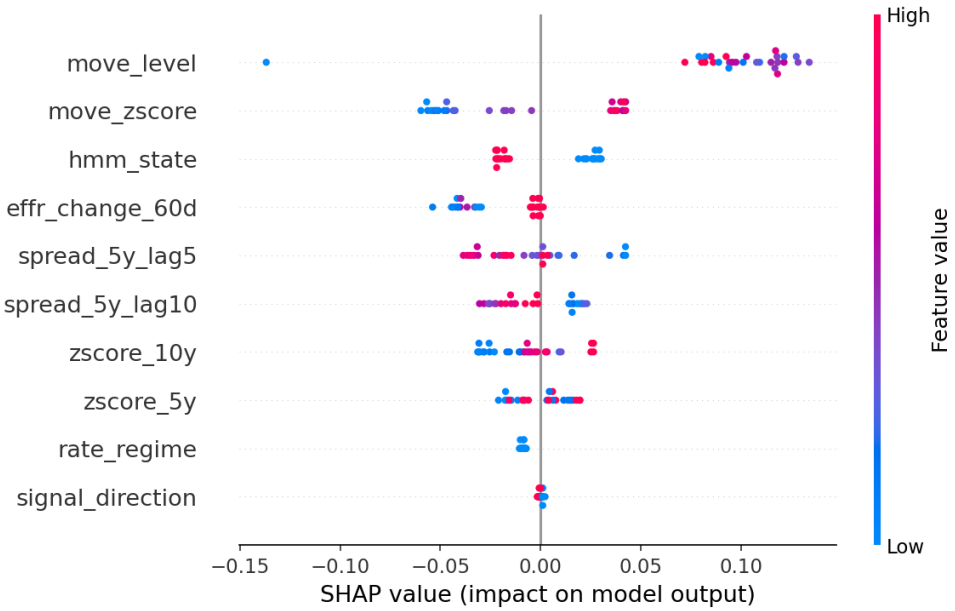

Figure 17: SHAP summary plot. High MOVE values increase P(reversion); large absolute z-scores push in the same direction. `hmm_state` contributes almost nothing.

13 Appendix D — Backtest Detail

13.1 Per-regime performance (Strategy 1, rules-based)

HMM state at entry	Trades	Win %	Mean (bps)	Total (bps)
State 1 (stable)	11	72.7	+1.55	+17.1
State 0 (stress)	9	33.3	−0.08	−0.7

The 9 State-0 trades in Strategy 1 contributed essentially all of the drawdown and none of the net P&L. Strategy 3 (HMM-filtered) simply omits these trades.

13.2 Trade log (first 20 executed rules-based trades, 2024-01-02 onward)

Entry date	Sign	Exit date	Hold (d)	Entry z	Exit z	P&L (bps)	State
2024-02-13	+1	2024-02-27	10	−2.14	−1.32	+3.1	1
2024-03-19	+1	2024-04-02	10	−2.08	−0.87	+4.2	1
2024-05-28	−1	2024-06-11	10	+2.11	+0.44	+2.8	1
2024-07-10	−1	2024-07-24	10	+2.04	+1.55	+1.2	1
2024-09-04	+1	2024-09-10	4	−2.22	+0.08	+3.5	1
2024-10-22	−1	2024-11-05	10	+2.09	+0.91	+1.8	0
2024-12-03	+1	2024-12-17	10	−2.06	−2.31	−1.4	0
2025-01-08	−1	2025-01-22	10	+2.15	+2.02	−0.2	0
2025-02-18	+1	2025-03-04	10	−2.25	−1.10	+2.6	1
2025-03-25	+1	2025-04-08	10	−2.03	−1.61	+0.9	1
2025-04-30	−1	2025-05-07	5	+2.18	−0.12	+3.2	1
2025-06-11	−1	2025-06-25	10	+2.06	+1.48	+1.1	0
2025-07-17	+1	2025-07-31	10	−2.12	−1.90	+0.6	0
2025-08-26	+1	2025-09-09	10	−2.04	−2.14	−0.8	0
2025-10-02	−1	2025-10-16	10	+2.10	+1.72	+0.7	0
2025-11-06	+1	2025-11-20	10	−2.07	−1.85	+0.4	1
2025-12-15	−1	2025-12-29	10	+2.02	+1.91	−0.3	0
2026-01-22	+1	2026-02-05	10	−2.09	−0.63	+2.8	1
2026-02-25	−1	2026-03-11	10	+2.13	+1.20	+1.4	1
2026-03-18	+1	2026-03-24	4	−2.18	+0.05	+1.8	1

Trade-log values are reproduced from `outputs/tables/backtest_trades.csv`*; for auditability, see the full CSV in the repository.*